Day 140 | Unit 10 | Teacher plan

AASA Applications and Cumulative Review

Course Final Test

Standards: 6.NS.B.3, 6.EE.A.1, 6.RP.A.1, 6.NS.C.5, 6.EE.A.2b, 6.SP.A.1, 6.NS.B.2, 6.EE.B.7, 6.RP.A.3a, 6.EE.A.2c, 6.G.A.1, 6.EE.A.2a, 6.EE.B.6, 6.NS.A.1, 6.RP.A.2, 6.RP.A.3b, 6.EE.A.3, 6.EE.A.4, 6.NS.B.4a, 6.SP.B.4, 6.SP.B.5c, 6.NS.C.7a, 6.NS.C.7b, 6.RP.A.3d, 6.NS.C.7c, 6.NS.C.7d, 6.RP.A.3c, 6.G.A.4, 6.EE.B.5, 6.NS.C.6b, 6.NS.C.6c, 6.EE.B.8, 6.G.A.2, 6.NS.C.8, 6.EE.C.9, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5d

Learning target: I can apply and explain Grade 6 mathematics across the full course.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

Administration

Use two test units of 24 questions. Allow flexible completion time and a break between units. Do not provide worked-example help or calculators. Keep the answer key separate. Every question is scored: 1 point per complete correct response, 48 total. Percentage = correct ÷ 48 × 100. These are classroom scores, not AASA scale scores.

Whole-course balance

Number System: 15 questions (31.25%); Expressions and Equations: 15 (31.25%); Ratios and Proportions: 10 (20.83%); Geometry: 4 (8.33%); Statistics and Probability: 4 (8.33%). The combined geometry/statistics share is 16.67%. These proportions fit the published Grade 6 AASA blueprint ranges.

Cognitive progression

Questions 1–8 establish foundations; 9–38 require application; 39–48 emphasize reasoning. These are instructional demand bands, not externally validated DOK ratings. The final samples the full course, so the previous-unit 20% rule applies only to the nine unit tests.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25

Course Final Test

Grade 6 Mathematics

Name: _____ Date: _____

Directions

This test has 48 questions in two test units of 24 questions. Work independently without a calculator. Read each question carefully. For multiple choice, select one answer; for multiple select, select every correct answer and no incorrect answer. Write exact numbers or expressions when requested, and label graph scales and units. Use the spaces provided for work.

This is original Math Coach practice modeled on public AASA item formats. It is not an official Arizona assessment.

Test Unit 1

Questions 1–24 | No calculator

01 / Constructed response

Use the standard algorithm: 39.5 + 8.2. Show your regrouping or division steps.

02 / Constructed response

Write 4^3 as repeated multiplication and evaluate.

03 / Constructed response

A tray has 5 red tiles and 5 blue tiles. Write the red-to-blue ratio in simplest form.

04 / Constructed response

A marker is 20 m below sea level. Write its signed elevation and explain what elevation 0 means.

05 / Constructed response

For 2x + 7y + 4, list terms, coefficients, and the constant.

06 / Multiple select

A class studies a collection of dogs. Select all statistical questions, and identify the expected variability.
A. How many centimeters are in 4 meters?
B. How does mass vary among the dogs?
C. What is 4 + 5?
D. What is the mass in kilograms of each of the dogs?

07 / Constructed response

Use long division: 300 ÷ 5. Show the algorithm and check.

08 / Constructed response

Solve x + 2 = 6. Check in the original equation.

09 / Multiple choice

Use the standard algorithm: 2.6 - 1.63. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 97/10
B. 97/100
C. 197/100
D. -3/100

10 / Multiple select

A recipe uses 3 cups oats for 2 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (6, 5)
B. (7, 4)
C. (6, 4)
D. (9, 6)

11 / Numeric entry and error analysis

Evaluate $3^2 + 4(5 - 3)$. Part A: enter the value. Part B: a student begins by adding 3^2 and 4; explain why that changes the expression.

12 / Multiple choice

Use the standard algorithm: 1.92 × 1.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 461/125
B. 211/125
C. 336/125
D. 672/25

13 / Multiple select

A triangle has base 14 cm and perpendicular height 5 cm. Select all expressions that give its area in cm².
A. 14 × 5
B. 7 × 5
C. (14 × 5) ÷ 2
D. (14 + 5) ÷ 2

14 / Expression entry and reasoning

A student buys x markers at $5 each and y folders at $2 each, then uses one $2 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

15 / Multiple choice

Use the standard algorithm: 13.75 ÷ 2.5. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 55
B. 9/2
C. 13/2
D. 11/2

16 / Double number line and explanation

Draw a double number line for 3 liters of paint covering 3 m². Part A: align 0, 1, 2, and 7 batches. Part B: enter the area covered by 21 liters and explain why the scales can be different.

17 / Numeric entry and reasoning

Let x = 3/2 and y = 3. Evaluate 7x + y². Part A: enter the value. Part B: show substitution and the order of operations.

18 / Multiple choice

Draw a bar partitioned into 4 equal parts. Shade 2 parts. Circle groups of 1 parts. Use the model to find 1/2 ÷ 1/4. Select one answer, then justify it with a model or calculation.
A. 3
B. 2
C. 1
D. 20

19 / Multiple select

5 notebooks cost $25. Select all statements at the same price per notebook.
A. 3 notebooks cost $16.
B. Each notebook costs $125.
C. 3 notebooks cost $15.
D. Each notebook costs $5.

20 / Multiple select

Select all expressions equivalent to 2(x + 4).
A. 2x + 4
B. 2x + 8
C. 8 + 2x
D. 6x

21 / Multiple choice

Calculate 7/4 ÷ 3/5. Simplify. Select one answer, then justify it with a model or calculation.
A. 12/35
B. 23/12
C. 47/12
D. 35/12

22 / Two-part numeric entry

A 10 m by 10 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

23 / Multiple select

Select all expressions equivalent to 5x + 6x + 5.
A. 16x
B. 11x + 6
C. (5 + 6)x + 5
D. 11x + 5

24 / Multiple choice

Calculate 5 ÷ 2 1/2. Simplify. Select one answer, then justify it with a model or calculation.
A. 2
B. 3
C. 1
D. 1/2

Test Unit 2

Questions 25–48 | No calculator

25 / Two-part numeric entry and reasoning

Store A sells 6 notebooks for $24. Store B sells 7 notebooks for $35. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

26 / Multiple select

Select all expressions equivalent to 6(x + 4).
A. 10x
B. 6x + 24
C. 6x + 4
D. 24 + 6x

27 / Multiple select

Select all true statements about 12 and 18. Explain why the greatest factor fits the kit situation.
A. 6 is the largest number of identical no-leftover kits from 12 pencils and 18 erasers.
B. 30 divides both counts.
C. 7 is their greatest common factor.
D. 6 divides both 12 and 18.

28 / Box plot and numeric entry

Six students recorded reading times (minutes): 5, 8, 8, 11, 11, 14. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

29 / Multiple select

Solve x - 5/2 = 2. Select every representation of the solution.
A. 18/4
B. 9/2
C. 11/2
D. 13/2

30 / Multiple select

A bicycle travels 40 km in 2 hours at constant speed. Select all true statements.
A. The speed is 42 km per hour.
B. In 2 hours it travels 40 km.
C. In 1 hour it travels 21 km.
D. The speed is 20 km per hour.

31 / Multiple select

Temperatures are -9 °C and -7 °C. Select all true comparisons.
A. -9 is warmer than -7.
B. -9 < -7
C. -7 lies to the right of -9 on a number line.
D. -9 > 0

32 / Multiple select

Solve 2 × x = 7. Select every representation of the solution.
A. 14/4
B. 9/2
C. 11/2
D. 7/2

33 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 7/2 feet.
A. 42 inches
B. 43 inches
C. 7/2 × 12 inches
D. 7/24 inches

34 / Multiple select

Consider -9 and 7. Select all true statements.
A. -9 is farther from zero than 7.
B. |-9| = -9
C. -9 > 7
D. |-9| = 9

35 / Multiple select

Solve x ÷ 3 = 13/6. Select every representation of the solution.
A. 26/4
B. 17/2
C. 13/2
D. 15/2

36 / Numeric entry and equation

A school club has 80 members; 15% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

37 / Two-part numeric entry

A closed box is 6 cm by 2 cm by 3 cm. Part A: enter its surface area. Part B: the 6-by-2 top is removed; enter the remaining surface area and explain.

38 / Data display and explanation

Reading-time frequencies (minutes) are 0–4: 3, 5–9: 4, 10–14: 5, 15–19: 4. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

39 / Model and constructed response

A jug contains 2 L. Each full bottle needs 4/3 L. How many full bottles can be filled, and how much liquid remains?

40 / Multiple select

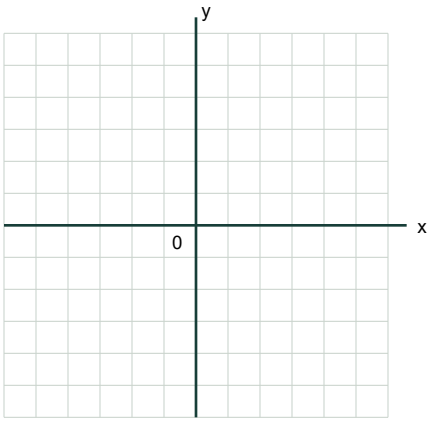
Select all values of x from the list that make x + 2 > 5 true. Explain why each rejected value fails.
A. 5
B. 3
C. 2
D. 4

41 / Multiple select

75 is 125% of an unknown quantity w. Select all equations that find w.
A. (125/100)w = 75
B. w = 75 + 125
C. w = 75 ÷ (125/100)
D. w = 75 × (125/100)

42 / Coordinate plotting and explanation

Plot P(-7, 4), Q(7, -4), and R(0, -4). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

43 / Multiple select

A club needs at least 4 members. Select all true statements for its membership count n.

A. $n \geq 4$

B. $n > 4$

C. 4 is an allowed membership count.

D. 3 is an allowed membership count.

44 / Numeric entry and model reasoning

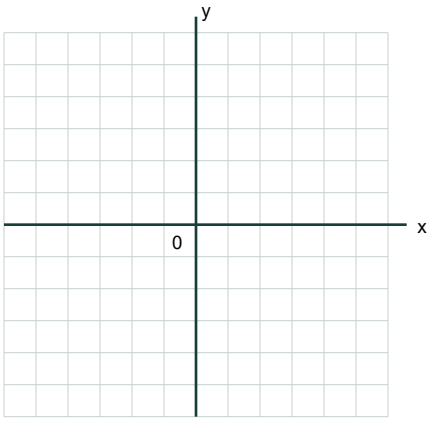
A prism measures 13/2 m by 5/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

45 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 20. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

46 / Coordinate plotting and numeric entry

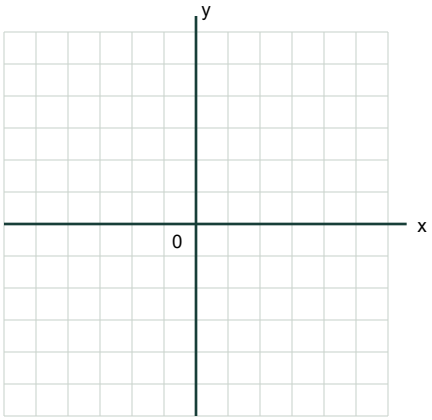
Plot A(-2, -3), B(3, -3), and C(-2, 3). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

47 / Table completion, equation, and graph

A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

48 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 2, 8, 8, 14, 14, 28. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

Day 140 | Unit 10 | Answer key

AASA Applications and Cumulative Review

Course Final Test

Standards: 6.NS.B.3, 6.EE.A.1, 6.RP.A.1, 6.NS.C.5, 6.EE.A.2b, 6.SP.A.1, 6.NS.B.2, 6.EE.B.7, 6.RP.A.3a, 6.EE.A.2c, 6.G.A.1, 6.EE.A.2a, 6.EE.B.6, 6.NS.A.1, 6.RP.A.2, 6.RP.A.3b, 6.EE.A.3, 6.EE.A.4, 6.NS.B.4a, 6.SP.B.4, 6.SP.B.5c, 6.NS.C.7a, 6.NS.C.7b, 6.RP.A.3d, 6.NS.C.7c, 6.NS.C.7d, 6.RP.A.3c, 6.G.A.4, 6.EE.B.5, 6.NS.C.6b, 6.NS.C.6c, 6.EE.B.8, 6.G.A.2, 6.NS.C.8, 6.EE.C.9, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5d

Learning target: I can apply and explain Grade 6 mathematics across the full course.

1. 47.7

Align decimal points; 39.5 + 8.2 = 47.7.

6.NS.B.3 | Constructed response

2. 4 × 4 × 4 = 64

The exponent counts 3 factors of 4; multiply all factors.

6.EE.A.1 | Constructed response

3. 1:1

Keep red first; divide 5:5 by 5.

6.RP.A.1 | Constructed response

4. -20 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

5. Terms 2x, 7y, 4; coefficients 2, 7; constant 4

Terms are separated by addition; coefficients multiply variables.

6.EE.A.2b | Constructed response

6. B, D - How does mass vary among the dogs?; What is the mass in kilograms of each of the dogs?

Different dogs can have different mass. The questions anticipate variation across a collection. The conversion and arithmetic questions have fixed answers.

6.SP.A.1 | Multiple select

7. 60

30 ÷ 5: write 6, subtract 30, remainder 0; bring down the next digit: 0 ÷ 5: write 0, subtract 0, remainder 0. Check: 5 × 60 = 300.

6.NS.B.2 | Constructed response

8. x = 4

Subtract 2 on both sides. Then x = 4; substitution gives 6 on each side.

6.EE.B.7 | Constructed response

9. B - 97/100

Align decimal points; 2.6 - 1.63 = 0.97.

6.NS.B.3 | Multiple choice

10. C, D - (6, 4); (9, 6)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

11. 17

Parentheses: 5 - 3 = 2. Exponent: $3^2 = 9$. Multiply: 4 × 2 = 8. Add: 9 + 8 = 17. The product must be completed before the final sum.

6.EE.A.1, 6.EE.A.2c | Numeric entry and error analysis

12. C - 336/125

Compute the whole-number product and restore decimal places: 1.92 × 1.4 = 2.688.

6.NS.B.3 | Multiple choice

13. B, C - 7 × 5; (14 × 5) ÷ 2

Triangle area is half the matching parallelogram: (14 × 5)/2 = 35 cm^2. Multiplying half the base by the height gives the same result.

6.G.A.1 | Multiple select

14. 5x + 2y - 2

Marker cost is 5x; folder cost is 2y. Add these subtotals, then subtract the one fixed coupon of 2. An expression does not need an equals sign.

6.EE.A.2a, 6.EE.B.6 | Expression entry and reasoning

15. D - 11/2

Multiply both numbers by 10: 137.5 ÷ 25 = 5.5. Check: 2.5 × 5.5 = 13.75.

6.NS.B.3 | Multiple choice

16. Aligned pairs (0,0), (3,3), (6,6), (21,21); 21 m^2

Each aligned jump adds 3 liters and 3 m^2. Both quantities scale by 7; the number lines measure different units.

6.RP.A.3a | Double number line and explanation

17. 39/2

Substitute: 7 × (3/2) + 3². The product is 21/2 and the square is 9. Add to get 39/2.

6.EE.A.2c | Numeric entry and reasoning

18. B - 2

Each small part is 1/4; 2 shaded parts divided into groups of 1 gives 2/1 = 2 groups.

6.NS.A.1 | Multiple choice

19. C, D - 3 notebooks cost $15.; Each notebook costs $5.

Cost per notebook = 25 ÷ 5 = 5. Multiply $5 by the chosen notebook count.

6.RP.A.2, 6.RP.A.3b | Multiple select

20. B, C - 2x + 8; 8 + 2x

Distribute 2 to both terms to get 2x + 8. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

21. D - 35/12

7/4 ÷ 3/5 = 7/4 × 5/3 = 35/12.

6.NS.A.1 | Multiple choice

22. Removed 12 m²; remaining 88 m²

Outer area = 100. Triangle = 4 × 6/2 = 12. Subtract the removed region exactly once.

6.G.A.1 | Two-part numeric entry

23. C, D - (5 + 6)x + 5; 11x + 5

Combine coefficients of like x terms: 5 + 6 = 11. The constant 5 stays separate.

6.EE.A.3 | Multiple select

24. A - 2

5 ÷ 5/2 = 5 × 2/5 = 2.

6.NS.A.1 | Multiple choice

25. A: $4 each; B: $5 each; A is the better buy.

Divide each cost by its own notebook count: 24/6 = 4; 35/7 = 5. Compare the same one-notebook unit.

6.RP.A.3b | Two-part numeric entry and reasoning

26. B, D - 6x + 24; 24 + 6x

Distribute 6 to both terms to get 6x + 24. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.4 | Multiple select

27. A, D - 6 is the largest number of identical no-leftover kits from 12 pencils and 18 erasers.; 6 divides both 12 and 18.

GCF(12, 18) = 6. Divide each count by 6: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

28. (5, 8, 19/2, 11, 14); IQR 3 min

Minimum and maximum anchor whiskers. Box ends are 8, 11; median line is 19/2. IQR = 11 - 8.

6.SP.B.4, 6.SP.B.5c | Box plot and numeric entry

29. A, B - 18/4; 9/2

Use the inverse operation on both sides. The solution is 9/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

30. B, D - In 2 hours it travels 40 km.; The speed is 20 km per hour.

Divide distance by time: 40 ÷ 2 = 20 km/h. Multiply this unit rate by 2 for the later distance.

6.RP.A.3b | Multiple select

31. B, C - -9 < -7; -7 lies to the right of -9 on a number line.

Among negatives, the number closer to zero is greater. Thus -7 is warmer and is farther right.

6.NS.C.7a, 6.NS.C.7b | Multiple select

32. A, D - 14/4; 7/2

Use the inverse operation on both sides. The solution is 7/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

33. A, C - 42 inches; 7/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 7/2 by 12, giving 42 inches. The physical length does not change.

6.RP.A.3d | Multiple select

34. A, D - -9 is farther from zero than 7.; |-9| = 9

Absolute value measures distance: 9 versus 7. A larger distance does not make the negative number greater.

6.NS.C.7c, 6.NS.C.7d | Multiple select

35. A, C - 26/4; 13/2

Use the inverse operation on both sides. The solution is 13/2; the doubled numerator and denominator name the same value. Substitution verifies the solution.

6.EE.B.7 | Multiple select

36. 12 participants; n = (15/100) × 80; less than the whole because the percent is below 100

Rename 15% as 3/20. Multiply by 80 to obtain 12. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

37. 72 cm²; open box 60 cm²

Closed area = 2(12+18+6) = 72. Subtract exactly one top face, 12, for the open box.

6.G.A.4 | Two-part numeric entry

38. 16 observations; highest intervals 10–14

Use four equal-width touching bars with heights 3, 4, 5, 4. Frequencies sum to 16. Include every interval with frequency 5.

6.SP.B.4 | Data display and explanation

39. 1 bottles; 2/3 L left

2 ÷ 4/3 = 2 × 3/4 = 3/2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

40. A, D - 5; 4

Subtract 2 conceptually: x must exceed 3. Equality at 3 fails because the inequality is strict.

6.EE.B.5 | Multiple select

41. A, C - (125/100)w = 75; w = 75 ÷ (125/100)

The part is percent rate times whole. Divide 75 by 5/4; w = 60. Check by multiplication.

6.RP.A.3c | Multiple select

42. P: II; Q: IV; R is on the y-axis.

P has signs (-,+), so II. Q has (+,-), so IV. R has x = 0 and is on an axis.

6.NS.C.6b, 6.NS.C.6c | Coordinate plotting and explanation

43. A, C - n ≥ 4; 4 is an allowed membership count.

At least includes the boundary, so n ≥ 4. On a real number line use a closed point and shade right; actual member counts are whole numbers.

6.EE.B.8 | Multiple select

44. 65/4 m³; 130 cubes

Multiply edges: 13/2 × 5/2 × 1 = 65/4. The edge counts are 13, 5, 2, giving 130 cubes. Each cube is 1/8 m³; 130/8 = 65/4.

6.G.A.2 | Numeric entry and model reasoning

45. 20%; denominator 100

Percent = (20 ÷ 100) × 100 = 20%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

46. AB = 5 units; AC = 6 units

AB is horizontal: 2 + 3 = 5. AC is vertical: 3 + 3 = 6. Each segment crosses an axis.

6.NS.C.8 | Coordinate plotting and numeric entry

47. (0,0), (1,4), (2,8); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

48. 6 observations; minutes; median 11, IQR 6; mean 37/3, MAD 19/3. A high value makes median/IQR a useful choice.

Order the observations. Middle positions give median 11 and half medians give IQR 6. Mean is 74/6; averaging distances gives MAD 19/3. Tie the chosen summary to the displayed shape and context.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d | Statistical reasoning and explanation